# Ashlee Hanif

**Lead Data Engineer | Data Architecture, Cloud Platforms, Big Data & Advanced Analytics**

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## Professional Summary

Results-driven Lead Data Engineer with 10+ years of experience designing, modernizing and optimizing enterprise data platforms across healthcare, technology and customer analytics environments. Proven expertise in cloud data architecture, lakehouse engineering and scalable ETL/ELT solutions across AWS, Azure and GCP. Highly skilled in Python, SQL, PySpark, Databricks, Snowflake and Apache Spark, delivering high-performance batch and real-time data processing solutions. Experienced in architecting event-driven pipelines, Kafka streaming, CDC integrations and distributed data processing frameworks for complex enterprise workloads. Strong background in Palantir Foundry, enterprise data modeling, dimensional warehousing and governed data products that support actionable business intelligence. Specialized in healthcare interoperability, EHR/EMR integration, HL7/FHIR standards and HIPAA-compliant data engineering with a focus on protecting sensitive information. Adept at establishing data governance, security, quality and lineage frameworks while improving platform reliability, performance and cost efficiency. Hands-on experience with DataOps, CI/CD, infrastructure automation and AI/ML-ready data ecosystems, enabling reliable analytics and intelligent applications. Recognized for technical leadership, mentoring engineering teams and translating complex business requirements into sustainable, high-impact data solutions.

## Technical Skills

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| **Programming & Scripting:** | Python, SQL, PySpark, Scala, Bash/Shell Scripting |
| **Cloud Platforms & Services:** | AWS (S3, Glue, EMR, Lambda, Kinesis, Step Functions, IAM), Microsoft Azure (Data Factory, ADLS Gen2, Synapse Analytics, Event Hubs, Key Vault), GCP (Cloud Storage, Dataflow, Pub/Sub) |
| **Data Lakehouse & Warehousing:** | Databricks, Delta Lake, Apache Iceberg, Snowflake, Amazon Redshift, Google BigQuery, Microsoft Fabric, OneLake, Medallion Architecture |
| **Big Data & Distributed Processing:** | Apache Spark, Spark SQL, Spark Structured Streaming, Hadoop, Hive, HDFS, MapReduce, Distributed Computing, Performance Tuning |
| **Data Integration & Streaming:** | Apache Kafka, Kafka Connect, Debezium, Change Data Capture (CDC), REST APIs, Real-Time Ingestion, Schema Evolution, Event-Driven Architecture |
| **ETL/ELT & Workflow Orchestration:** | Apache Airflow, dbt Core/Cloud, Databricks Workflows, SSIS, Informatica IICS, Incremental Processing, Pipeline Testing |
| **Palantir Foundry & Healthcare Integration:** | Palantir Foundry (Pipeline Builder, Code Repositories, Ontology, Workshop); Healthcare (EHR/EMR, Epic Clarity, HL7, FHIR, ADT, Claims, Eligibility, Pharmacy Data, Patient 360) |
| **Data Modeling & Architecture:** | Kimball Dimensional Modeling, Star/Snowflake Schemas, Data Vault, Data Mesh, SCD Type 1/2, Data Contracts, Enterprise Data Architecture |
| **Data Governance, Security & Compliance:** | HIPAA, HITRUST, PHI/PII Protection, Unity Catalog, RBAC, Encryption, Data Masking, Data Cataloging, Data Quality, Data Lineage |
| **DevOps, DataOps & CI/CD:** | Terraform, Docker, Kubernetes, GitHub Actions, Azure DevOps, Jenkins, Git, CI/CD, Infrastructure as Code (IaC) |
| **Data Reliability & Observability:** | Datadog, Prometheus, Grafana, CloudWatch, Automated Validation, SLA/SLO Monitoring, Root Cause Analysis, Incident Management, Pipeline Observability |
| **AI/ML, Databases & Business Intelligence:** | MLflow, Feature Engineering, Feature Stores, Vector Databases, Retrieval-Augmented Generation (RAG), AI-Ready Data Pipelines, PostgreSQL, SQL Server, Oracle, MySQL, MongoDB, Power BI, Tableau |

## Professional Experience

### Lead Data Engineer

**Company:** BlueConic **Date:** July 2023 – Present

- Architected and led the implementation of enterprise-scale Databricks Lakehouse solutions across AWS and Azure, integrating **35+ heterogeneous data sources** through Delta Lake, Medallion Architecture and reusable ingestion frameworks to process **8+ TB of daily customer, transactional and behavioral data** while improving scalability and analytical accessibility.
- Spearheaded the development of high-throughput customer data engineering platforms using Python, PySpark, Apache Kafka and Spark Structured Streaming, processing **15M+ daily events** from CRM systems, digital applications and

customer engagement platforms to support real-time segmentation, personalization and advanced customer intelligence.

- Designed and optimized enterprise CDC and event-driven ingestion architectures using Debezium, Kafka Connect, Delta Lake and incremental processing strategies, reducing end-to-end data latency by **65%** while strengthening fault tolerance, schema evolution and reliable delivery across mission-critical analytical workloads.
- Engineered governed operational data products using Palantir Foundry, leveraging Pipeline Builder, Code Repositories, Ontology and Workshop to transform complex enterprise datasets into reusable business entities, establish consistent relationships across operational systems and accelerate business-facing analytics delivery by **40%**.
- Established enterprise-wide data governance and security frameworks using Databricks Unity Catalog, role-based access controls, automated lineage, data contracts and PII protection policies, standardizing access management and improving data quality, traceability and compliance readiness across **180+ enterprise datasets**.
- Directed the modernization of **70+ legacy ETL/ELT workflows** using Apache Airflow, dbt, Databricks Workflows, Terraform and automated CI/CD pipelines, implementing incremental transformations, orchestration standards and workload optimization strategies that reduced infrastructure costs by **35%** and improved production pipeline reliability to **99.9%**.
- Provided technical leadership to a team of **12 data engineers**, driving architecture reviews, engineering standards, Spark performance optimization, DataOps observability and MLflow-enabled data workflows while mentoring engineers, resolving complex production challenges and improving deployment efficiency by **45%**.

## Senior Data Engineer

**Company:** Sunflower Lab **Date:** Oct 2019 – June 2023

- Designed and engineered HIPAA-compliant healthcare data pipelines using AWS Glue, Amazon S3, Snowflake, Python and PySpark, integrating **25+ EHR/EMR, insurance claims, eligibility, pharmacy and clinical data sources** into secure analytical environments to support healthcare reporting, operational intelligence and population health initiatives.
- Developed scalable healthcare interoperability and data integration frameworks using HL7, FHIR, ADT messages and REST APIs, standardizing clinical and administrative data from disparate healthcare applications while consolidating **12M+ patient records** into trusted Patient 360 datasets for care coordination and longitudinal analytics.
- Built and optimized enterprise transformation frameworks using dbt, Apache Airflow and Snowflake, developing reusable SQL models, incremental loading strategies, SCD Type 2 transformations and dimensional schemas to improve analytical consistency, automate complex healthcare reporting and reduce processing time by **45%**.
- Implemented comprehensive healthcare data security and governance controls including PHI/PII de-identification, encryption, RBAC, secure data access policies and audit logging, strengthening HIPAA-aligned data handling practices while protecting sensitive patient information across **100+ regulated datasets**.
- Optimized distributed Spark processing and Snowflake analytical workloads through partitioning, query profiling, warehouse tuning, incremental CDC ingestion and resource optimization, improving query execution performance by **50%** while reducing cloud compute consumption by **30%** across high-volume healthcare datasets.
- Strengthened production data platform reliability and deployment automation using Terraform, GitHub Actions, automated data quality testing and Datadog monitoring, introducing validation checkpoints, exception handling and proactive alerting that supported **99.8% pipeline availability** across critical healthcare workflows.

## Data Engineer

**Company:** Opallios Inc **Date:** Mar 2017 – Sep 2019

- Developed and maintained scalable cloud-native ETL/ELT pipelines using AWS S3, AWS Glue, Amazon Redshift, Python and PySpark, integrating **20+ enterprise applications** and processing **2+ TB of daily operational data** to support centralized reporting, business intelligence and downstream analytical workloads.
- Engineered reusable Apache Spark and SQL-based transformation frameworks to process structured and semi-structured datasets, implementing partitioning, incremental ingestion, schema evolution and optimized data storage strategies that reduced batch processing time by **40%** and improved pipeline maintainability.
- Designed and implemented enterprise analytical data models using Kimball dimensional modeling, star schemas, fact and dimension tables and SCD Type 1/2 techniques, transforming complex operational data into consistent reporting structures and improving analytical accuracy across financial and business operations.
- Automated data ingestion and workflow orchestration using Apache Airflow, AWS Lambda and Step Functions, developing dependency-driven schedules, retry mechanisms and exception-handling workflows that reduced manual intervention by **55%** and improved the consistency of scheduled data delivery.
- Implemented automated data validation, reconciliation and quality monitoring frameworks using Python, SQL and rule-based anomaly detection, identifying source-to-target discrepancies, enforcing transformation accuracy and reducing recurring production data quality incidents by **35%**.

## ETL Developer

**Company:** Datagration **Date:** Aug 2014 – Feb 2017

- Developed and maintained enterprise ETL workflows using SQL Server, SSIS, Python and Apache Hadoop, extracting, cleansing and integrating structured and semi-structured datasets from multiple operational systems into centralized data repositories for reporting and downstream analytics.
- Built and optimized distributed batch processing solutions using Apache Spark, Hive, HDFS and MapReduce, implementing parallel data transformations, efficient file processing and optimized resource allocation to improve large-scale data processing throughput by **40%**.
- Enhanced relational database performance and query execution efficiency by optimizing complex SQL statements, stored procedures, joins, indexing strategies and execution plans, reducing processing time by **45%** and improving responsiveness across operational reporting environments.
- Automated recurring data extraction, cleansing and migration activities using Python, Shell scripting and reusable ETL components, implementing scheduled batch execution, transformation logic and error-handling procedures that reduced manual processing effort by **50%**.
- Implemented source-to-target mappings, reconciliation procedures and data quality controls across **30+ enterprise integration workflows**, supporting database migrations, resolving production data inconsistencies and improving the accuracy and reliability of downstream reporting datasets.

## Projects

### Customer Data Platform & Real-Time Analytics

**Core Technologies:** Python, PySpark, SQL, Databricks, Delta Lake, Apache Kafka, Kafka Connect, Debezium, Spark Structured Streaming, Palantir Foundry, AWS, Azure

- Architected a unified customer data platform integrating **35+ CRM, transactional and digital engagement sources** across AWS and Azure, processing **8+ TB of daily data** through scalable Databricks Lakehouse architecture.
- Developed high-throughput ingestion and streaming workflows using Python, PySpark and Apache Kafka to process **15M+ daily customer events** for behavioral analytics and personalized engagement.
- Implemented CDC pipelines using Debezium and Kafka Connect, reducing end-to-end data latency by **65%** while improving real-time data availability.
- Established governed data products through Palantir Foundry, Databricks Unity Catalog and automated lineage to strengthen enterprise data accessibility and reliability.

### Healthcare Data Integration & Patient 360 Analytics

**Core Technologies:** Python, PySpark, SQL, AWS Glue, Amazon S3, Snowflake, dbt, Apache Airflow, HL7, FHIR, ADT, REST APIs, HIPAA

- Designed HIPAA-compliant healthcare data integration pipelines connecting **25+ EHR/EMR, clinical, claims, eligibility and pharmacy sources** into centralized analytical environments.
- Developed interoperability workflows using HL7, FHIR, ADT messages and REST APIs, standardizing clinical information and consolidating **12M+ patient records** into longitudinal Patient 360 datasets.
- Implemented PHI/PII de-identification, encryption, role-based access controls and audit logging to protect sensitive patient information across **100+ regulated datasets**.
- Optimized Snowflake transformations and distributed Spark workloads, improving query performance by **50%** and reducing processing time by **45%**.

### AWS Data Integration & Operational Analytics

**Core Technologies:** Python, PySpark, SQL, AWS S3, AWS Glue, Amazon Redshift, Apache Spark, Apache Airflow, AWS Lambda, Step Functions, Kimball Dimensional Modeling, SCD Type 1/2

- Developed scalable cloud-native data integration frameworks using AWS to consolidate structured and semi-structured information from **20+ enterprise applications** into centralized analytical repositories.
- Engineered distributed transformation workflows using Apache Spark, PySpark and SQL, processing **2+ TB of daily operational data** through incremental ingestion, partitioning and schema evolution.
- Designed dimensional data models using Kimball methodology and SCD Type 1/2 to improve reporting consistency and downstream analytical performance.
- Automated pipeline orchestration through Apache Airflow, AWS Lambda and Step Functions, reducing batch processing time by **40%** and manual intervention by **55%**.

## Education

**Community College of Philadelphia**

Bachelor of Science in Computer Science